

Essential Summary

A Modified Generalized Harmonic Function Perturbation Method and Its Application in Analyzing Generalized Duffing–Harmonic –Rayleigh–Liénard Oscillator

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1 Background and Motivation

This is the **foundational paper** of the MGHFP method series. Many classical analytical methods can analyze nonlinear oscillators quantitatively — but only when the restoring force and damping are simple. Once the oscillator becomes complicated, these methods **cannot execute their procedures symbolically** without first assigning numerical values to all system parameters. They can only produce isolated solution snapshots, never a continuous analytical picture of how solutions evolve with changing parameters.

This work introduces the **modified generalized harmonic function perturbation (MGHFP) method** by embedding the composite Simpson quadrature formula into the classical GHFP framework, enabling the entire perturbation procedure to proceed purely symbolically — no parameter pre-assignment required. This is the foundation upon which all subsequent MGHFP applications (IJNLM 2025, Phys. Scr. 2026) are built.

2 System Model

The generalized Duffing–Harmonic–Rayleigh–Liénard (DHRL) oscillator (Eq. 1 from the original paper):

$$\ddot{x} + \frac{\lambda x + \mu x^3}{1 + \nu x^2} = \varepsilon(\mu_c + \mu_2 x^2 + \mu_4 x^4 + \mu_6 x^6 + \mu_{22} \dot{x}^2 + \mu_{24} \dot{x}^4) \dot{x} \quad (1)$$

where λ, μ, ν are stiffness parameters, and the right side contains a **6-term generalized nonlinear damping** combining Rayleigh terms ($\dot{x}^2, \dot{x}^4$) and Liénard terms (x^2, x^4, x^6). The rational restoring force makes this oscillator a maximally challenging test case.

3 The MGHFP Method

3.1 How It Works

The classical GHFP method proceeds via: (1) nonlinear time transformation $\frac{d\varphi}{dt} = \Phi(\varphi)$, (2) Fourier expansion of the solution $x = a \cos^2 \Phi + b$, (3) perturbation expansion $a = a_0 + \varepsilon a_1 + \dots$, and (4) evaluation of integrals to obtain Fourier coefficients. Step (4) is the bottleneck — for complicated oscillators, these integrals **cannot be evaluated analytically**.

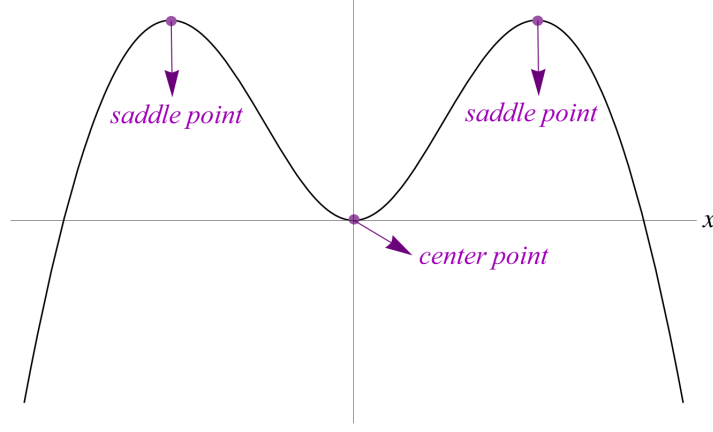


Figure 1: **Figure 6.** Potential portrait of the DHRL oscillator. The rich potential landscape enables homoclinic and heteroclinic bifurcations, making this an ideal testbed for the MGHFP method.

3.2 The MGHFP Fix — Composite Simpson Integration

The MGHFP method discretizes the integrals using the composite Simpson quadrature formula:

$$\int_a^b f(x) dx \approx \frac{h}{3} \left[f(x_0) + 4 \sum_{i \text{ odd}} f(x_i) + 2 \sum_{i \text{ even}} f(x_i) + f(x_n) \right] \quad (2)$$

By dividing the integration interval into eight equal subintervals, the Fourier coefficients p_{2i} become **explicit algebraic functions** of a_0 , b , and system parameters. The entire perturbation procedure now runs purely symbolically.

3.3 Nonlinear Time Transformation Details

The solution is assumed as $x(t) = a \cos^2 \varphi(t) + b$, with the nonlinear time transformation:

$$\frac{d\varphi}{dt} = \Phi(\varphi) = \Phi_0 + \varepsilon \Phi_1 + \dots \quad (3)$$

The zero-order frequency function is:

$$\Phi_0(\varphi) = \frac{\sqrt{2[V(a_0 + b) - V(a_0 \cos^2 \varphi + b)]}}{x'_0(\varphi)} \quad (4)$$

where $V(x) = \int g(x) dx$ is the potential energy. For complicated $g(x)$, this expression is analytically intractable — which is precisely why the Simpson discretization is needed. After discretization, Φ_0 is expanded as a Fourier series $\Phi_0(\varphi) = \sum_{i=0}^M (p_{2i} \cos 2i\varphi + q_{2i} \sin 2i\varphi)$ with $M = 4$.

4 Key Equations

4.1 Amplitude–Parameter Relation (Central Result)

The analytical relationship between limit cycle amplitude A and control parameter μ_c :

$$\mu_c = -\frac{4}{a_0^2(2p_0 - p_4)} \left(E_2\mu_2 + E_4\mu_4 + E_6\mu_6 + E_{22}\mu_{22} + E_{24}\mu_{24} \right) \quad (5)$$

where E_i are functions of a_0 , b , and the Fourier coefficients p_{2i} computed via composite Simpson integration. Given system parameters, the limit cycle amplitude is obtained **analytically** — no numerical integration needed.

4.2 Stability Criterion

$$h_0 = \frac{1}{\pi} \int_0^\pi \left[\mu_c + \sum \mu_i (a_0 \cos^2 \varphi + b)^i + \sum \mu_{ij} (a_0 \cos^2 \varphi + b)^i (\dot{x}_0)^2 \right] d\varphi \quad (6)$$

$h_0 > 0$: unstable; $h_0 = 0$: semi-stable (bifurcation point); $h_0 < 0$: stable.

4.3 Homoclinic and Heteroclinic Bifurcation Thresholds

Setting the limit cycle amplitude equal to the saddle point coordinate in Eq. (5) yields the critical μ_c^{hom} for homoclinic bifurcation. For heteroclinic bifurcation, the amplitude is set to the coordinates of both connected saddle points, yielding critical μ_c^{hetero} values. The variation of μ_c^{hetero} with respect to system parameters is analyzed in Figure 8.

5 Example 1: Limit Cycle Evolution (Section 4.2)

The following parameters are used: $\lambda = 2$, $\mu = 1$, $\nu = 5$, $\mu_2 = -0.88$, $\mu_4 = 0.39$, $\mu_6 = -0.1$, $\mu_{22} = 0.17$, $\mu_{24} = 0.18$.

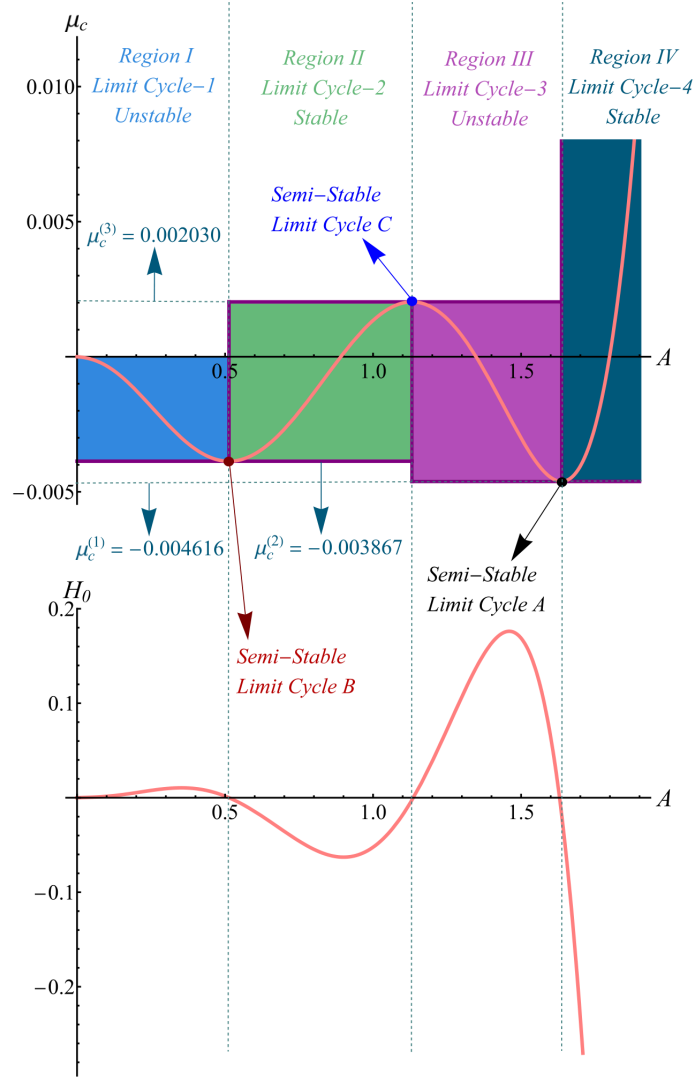


Figure 2: **Figure 2.** Evolution of limit cycles in the parametric plane (μ_c - A curve). Four regions (I–IV) identified. Semi-stable bifurcation points A and B mark transitions between regions.

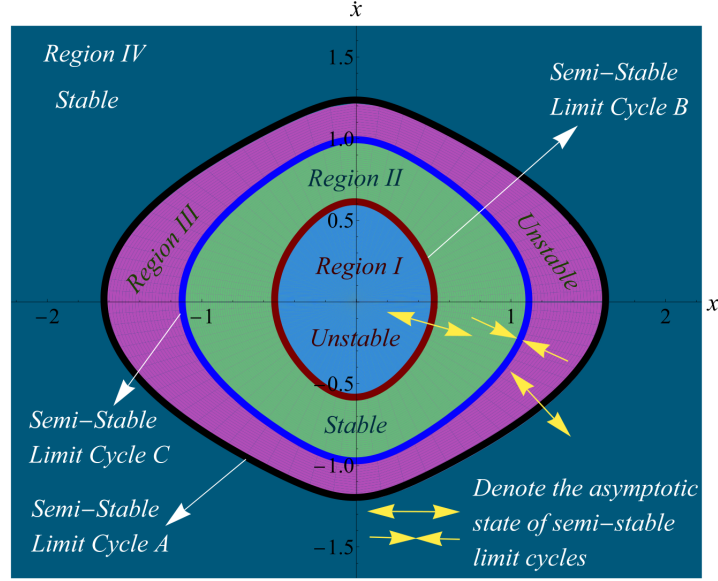


Figure 3: **Figure 3.** Phase portraits corresponding to the four regions in Figure 2. The method accurately captures transitions between stability regimes.

Figure 2 reveals the complete limit cycle lifecycle: Region I (no limit cycle), Point A (semi-stable emergence), Region II (stable+unstable coexistence), Point B (second semi-stable bifurcation), Region III (one stable+one unstable), Region IV (single stable cycle persists). Figure 3 provides corresponding phase portraits.

6 Example 2: Heteroclinic Bifurcation (Section 4.3)

The heteroclinic bifurcation analysis focuses on deriving **analytical expressions** for μ_c^{hetero} , enabling quantitative study of its variation with system parameters.

7 Key Findings

1. **First purely symbolic GHFP variant:** The introduction of composite Simpson quadrature into the GHFP framework makes the proposed method the first variant capable of **purely symbolic execution** for complicated oscillators — no parameter pre-assignment needed. This is the foundational innovation enabling all subsequent MGHFP work.
2. **Complete lifecycle from parametric plane (Figures 2–3):** The μ_c – A curve reveals the entire lifecycle of each limit cycle, with four distinct regions separated by semi-stable bifurcation points. Phase portraits in each region visually confirm the parametric analysis.
3. **Homoclinic and heteroclinic bifurcations predicted (Figures 8–9):** Both types of global bifurcation are quantitatively analyzed. The analytical expression for μ_c^{hetero} enables systematic study of its dependence on all damping coefficients simultaneously.
4. **Method validated on maximally challenging test case:** The DHRL oscillator combines rational restoring force $\frac{\lambda x + \mu x^3}{1 + \nu x^2}$ with six-term generalized Rayleigh–Liénard damping. All analytical predictions agree with Runge–Kutta results.
5. **Foundation for the MGHFP series:** This paper establishes the framework subsequently extended to SD oscillators with quartic damping (IJNLM 2025, with Padé enhancement) and coupled SD oscillators with two irrational terms (Phys. Scr. 2026).

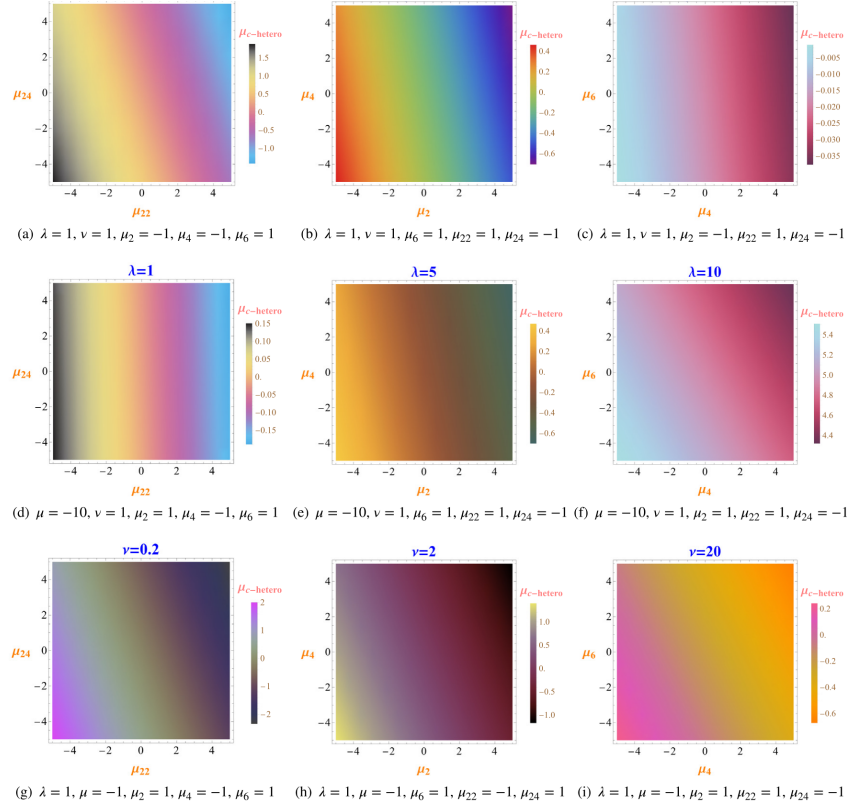


Figure 4: **Figure 8.** Variation of the heteroclinic bifurcation parameter μ_c^{hetero} with system parameters in different conditions. Each subfigure shows how μ_c^{hetero} changes as one damping coefficient is varied while others are fixed.

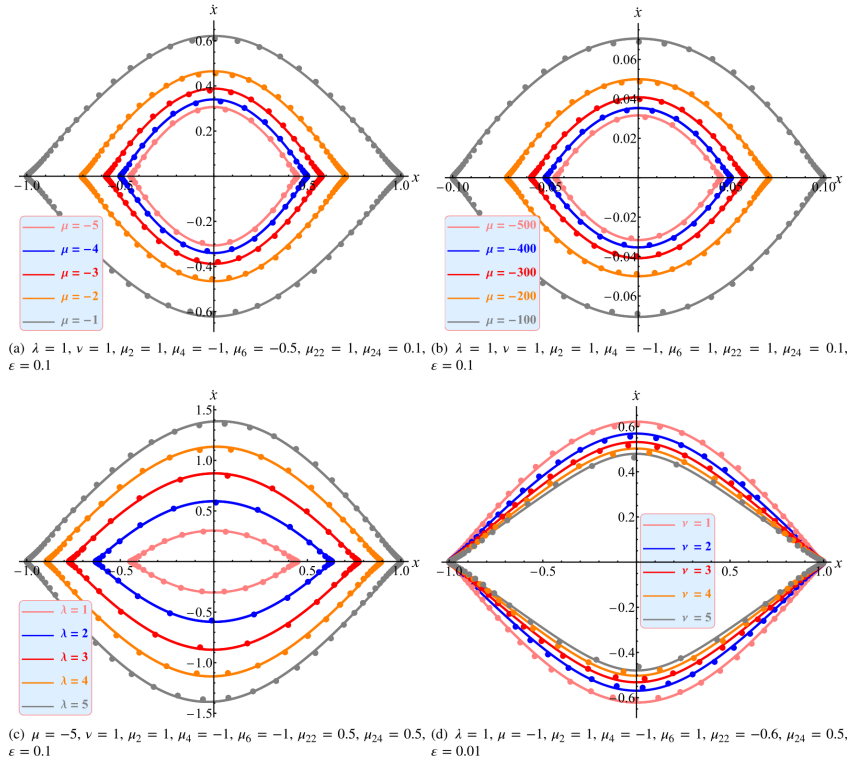


Figure 5: **Figure 9.** Phase portraits of heteroclinic solutions with different parameters. Each subfigure compares the analytical MGHFP solution with the Runge–Kutta numerical reference, validating the method's accuracy for heteroclinic orbit prediction.

8 Engineering Significance

The generalized DHRL oscillator models realistic scenarios in MEMS resonators (squeeze-film damping), vibration isolators (nonlinear electromagnetic damping), and energy harvesting (self-excited systems). The MGHFP method provides the first analytical tool capable of producing continuous amplitude-parameter curves for such systems.

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